

Narrower Is Not Automatically Better

Topic 6.2 · TODAY'S PROBLEM

A library system has 20,000 adult cardholders and wants to estimate p , the proportion who used a digital audiobook last month.

Plan A: A computer takes an SRS of 200 cardholder IDs. All answer; 116 say yes.

Plan B: A newsletter link lets cardholders choose to respond once. Of 800 respondents, 520 say yes.

Eli: “Use Plan B. Its larger sample gives the narrower interval.”

Dana: “Use Plan A. Entry into the sample matters more than the narrowest output.”

Plan counts

Plan	N	n	Yes	No
A	20,000	200	116	84
B	20,000	800	520	280

FIRST TAKE (5 min, on your sheet)

Which plan, if either, should estimate all adult cardholders? Cite one collection detail.

- DOK 1** (a) Define the population parameter p in context and name the confidence-interval procedure that would be appropriate when its conditions are met.
- DOK 2** (b) For each plan, compute $\hat{p}$, check the random, 10 percent, and large-count conditions, and calculate the 95 percent formula interval. Mark any nondefensible population interval.
- DOK 3** (c) Decide which plan can support a 95 percent interval for all cardholders. Cite recruitment and a numerical condition, compare interval widths, and explain the reader consequence of reporting Plan B's output.

Video + follow-along: u6_lesson1-2_live.html (scan the code)

Finish (a)–(c) after the video. Turn in your sheet.

